

No SQL Project
On
E-Commerce Product & Review
Management System Using MongoDB

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Batch: BCA DS-23

Department: BCA (DS/AI)

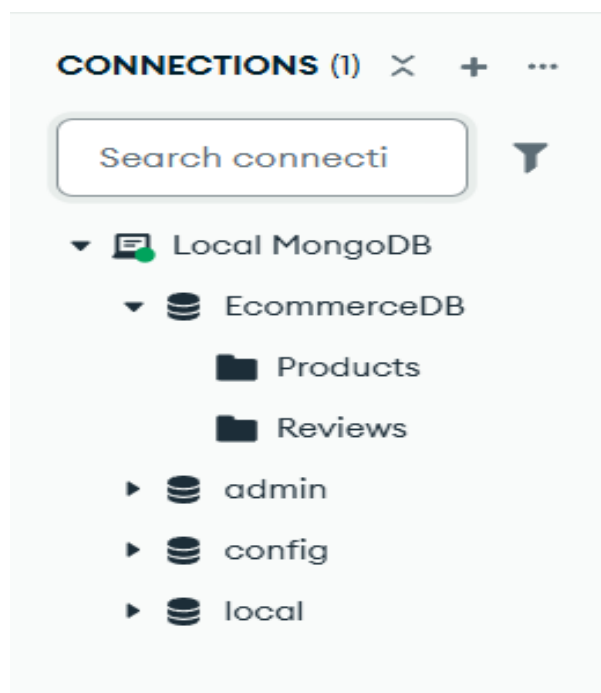
1.CREATE DATABASE

MongoDB Query: use EcommerceDB

```
>_MONGOSH  
  
> use("EcommerceDB")  
< switched to db EcommerceDB
```

2.CREATE COLLECTION

MongoDB Query: db.createCollection("Products")



3.INSERT 30 PRODUCTS

MongoDB Query:

db.Products.insertMany([

{ product_id: "P001", name: "Wireless Bluetooth Headphones",
category: "Electronics", brand: "Sony", price: 2499, rating: 4.5, stock:
25, seller: "TechStore" },

{ product_id: "P002", name: "Smart Watch", category: "Electronics",
brand: "boAt", price: 1999, rating: 4.2, stock: 40, seller: "GadgetWorld"
},

{ product_id: "P003", name: "Running Shoes", category: "Footwear",
brand: "Nike", price: 3499, rating: 4.6, stock: 18, seller: "SportsHub" },

{ product_id: "P004", name: "Face Moisturizer", category: "Beauty",
brand: "Cetaphil", price: 599, rating: 4.4, stock: 50, seller: "BeautyMart"
},

{ product_id: "P005", name: "Laptop Backpack", category:
"Accessories", brand: "American Tourister", price: 1299, rating: 4.3,
stock: 30, seller: "BagWorld" },

{ product_id: "P006", name: "Mechanical Keyboard", category:
"Electronics", brand: "Redragon", price: 2899, rating: 4.5, stock: 15,
seller: "TechStore" },

{ product_id: "P007", name: "USB-C Fast Charger", category:
"Electronics", brand: "Samsung", price: 1299, rating: 4.4, stock: 45,
seller: "MobileZone" },

{ product_id: "P008", name: "Denim Jacket", category: "Fashion",
brand: "Levi's", price: 2799, rating: 4.5, stock: 20, seller: "FashionHub" },

{ product_id: "P009", name: "Stainless Steel Water Bottle", category:
"Home & Kitchen", brand: "Milton", price: 699, rating: 4.3, stock: 60,
seller: "HomeStore" },

{ product_id: "P010", name: "Portable Bluetooth Speaker", category: "Electronics", brand: "JBL", price: 3299, rating: 4.7, stock: 22, seller: "SoundWorld" },

{ product_id: "P011", name: "Cotton T-Shirt", category: "Fashion", brand: "H&M", price: 799, rating: 4.1, stock: 75, seller: "FashionHub" },

{ product_id: "P012", name: "Coffee Maker", category: "Home & Kitchen", brand: "Philips", price: 3499, rating: 4.4, stock: 12, seller: "HomeStore" },

{ product_id: "P013", name: "Gaming Mouse", category: "Electronics", brand: "Logitech", price: 1599, rating: 4.6, stock: 35, seller: "GameZone" },

{ product_id: "P014", name: "Sunscreen SPF 50", category: "Beauty", brand: "Minimalist", price: 499, rating: 4.5, stock: 80, seller: "BeautyMart" },

{ product_id: "P015", name: "Yoga Mat", category: "Sports", brand: "Boldfit", price: 899, rating: 4.2, stock: 40, seller: "SportsHub" },

{ product_id: "P016", name: "Air Fryer", category: "Home & Kitchen", brand: "Philips", price: 6999, rating: 4.6, stock: 10, seller: "HomeStore" },

{ product_id: "P017", name: "Smartphone", category: "Electronics", brand: "OnePlus", price: 29999, rating: 4.5, stock: 14, seller: "MobileZone" },

{ product_id: "P018", name: "Casual Handbag", category: "Accessories", brand: "Lavie", price: 1799, rating: 4.2, stock: 28, seller: "BagWorld" },

{ product_id: "P019", name: "LED Desk Lamp", category: "Home & Kitchen", brand: "Wipro", price: 1099, rating: 4.3, stock: 32, seller: "HomeStore" },

{ product_id: "P020", name: "Gaming Headset", category: "Electronics", brand: "HyperX", price: 4999, rating: 4.7, stock: 16, seller: "GameZone" },

{ product_id: "P021", name: "Skin Care Gift Set", category: "Beauty", brand: "The Body Shop", price: 2199, rating: 4.4, stock: 24, seller: "BeautyMart" },

{ product_id: "P022", name: "Formal Shirt", category: "Fashion", brand: "Van Heusen", price: 1599, rating: 4.1, stock: 35, seller: "FashionHub" },

{ product_id: "P023", name: "Cricket Bat", category: "Sports", brand: "SG", price: 2499, rating: 4.5, stock: 17, seller: "SportsHub" },

{ product_id: "P024", name: "Wireless Mouse", category: "Electronics", brand: "HP", price: 899, rating: 4.2, stock: 55, seller: "TechStore" },

{ product_id: "P025", name: "Electric Kettle", category: "Home & Kitchen", brand: "Prestige", price: 1499, rating: 4.3, stock: 26, seller: "HomeStore" },

{ product_id: "P026", name: "Sunglasses", category: "Accessories", brand: "Fastrack", price: 1299, rating: 4.4, stock: 38, seller: "FashionHub" },

{ product_id: "P027", name: "Bluetooth Earbuds", category: "Electronics", brand: "Realme", price: 1799, rating: 4.3, stock: 42, seller: "MobileZone" },

{ product_id: "P028", name: "Backpack", category: "Accessories", brand: "Wildcraft", price: 1599, rating: 4.5, stock: 31, seller: "BagWorld" },

{ product_id: "P029", name: "Foundation Makeup", category: "Beauty", brand: "Maybelline", price: 749, rating: 4.2, stock: 48, seller: "BeautyMart" },

{ product_id: "P030", name: "Resistance Bands Set", category: "Sports", brand: "Decathlon", price: 699, rating: 4.4, stock: 65, seller: "SportsHub" }

])

MongoDB Compass - Local MongoDB/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

Local MongoDB

EcommerceDB

Products

admin

config

local

Welcome mongosh: Local Mongo...

> MONGOSH

< switched to db EcommerceDB

> db.Products.insertMany([

```
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{product_id: "P028", name: "Backpack", category: "Accessories", brand: "Wildcraft", price: 1599, rating: 4.5, stock: 31, seller: "BagWorld"},
{product_id: "P029", name: "Foundation Makeup", category: "Beauty", brand: "Maybelline", price: 749, rating: 4.2, stock: 48, seller: "BeautyMart"},
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])
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MongoDB Compass - Local MongoDB/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

Local MongoDB

EcommerceDB

Products

admin

config

local

Welcome mongosh: Local Mongo...

> MONGOSH

```
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  {product_id: "P020", name: "Gaming Headset", category: "Electronics", brand: "HyperX", price: 4999, rating: 4.7, stock: 16, seller: "GameZone"},
  {product_id: "P021", name: "Skin Care Gift Set", category: "Beauty", brand: "The Body Shop", price: 2199, rating: 4.4, stock: 24, seller: "BeautyMart"},
  {product_id: "P022", name: "Formal Shirt", category: "Fashion", brand: "Van Heusen", price: 1599, rating: 4.1, stock: 35, seller: "FashionHub"},
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  {product_id: "P027", name: "Bluetooth Earbuds", category: "Electronics", brand: "Realme", price: 1799, rating: 4.3, stock: 42, seller: "MobileZone"},
  {product_id: "P028", name: "Backpack", category: "Accessories", brand: "Wildcraft", price: 1599, rating: 4.5, stock: 31, seller: "BagWorld"},
  {product_id: "P029", name: "Foundation Makeup", category: "Beauty", brand: "Maybelline", price: 749, rating: 4.2, stock: 48, seller: "BeautyMart"},
  {product_id: "P030", name: "Resistance Bands Set", category: "Sports", brand: "Decathlon", price: 699, rating: 4.4, stock: 65, seller: "SportsHub"}
])
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 insertedIds: {
   '0': ObjectId('6ab361d683d0a67233feaa46'),
   '1': ObjectId('6ab361d683d0a67233feaa47'),
   '2': ObjectId('6ab361d683d0a67233feaa48'),
   '3': ObjectId('6ab361d683d0a67233feaa49'),
   '4': ObjectId('6ab361d683d0a67233feaa4a'),
   '5': ObjectId('6ab361d683d0a67233feaa4b'),
   '6': ObjectId('6ab361d683d0a67233feaa4c'),
   '7': ObjectId('6ab361d683d0a67233feaa4d'),
   '8': ObjectId('6ab361d683d0a67233feaa4e'),
   '9': ObjectId('6ab361d683d0a67233feaa4f'),
   '10': ObjectId('6ab361d683d0a67233feaa50'),
   '11': ObjectId('6ab361d683d0a67233feaa51'),
   '12': ObjectId('6ab361d683d0a67233feaa52'),
   '13': ObjectId('6ab361d683d0a67233feaa53'),
   '14': ObjectId('6ab361d683d0a67233feaa54'),
   '15': ObjectId('6ab361d683d0a67233feaa55'),
   '16': ObjectId('6ab361d683d0a67233feaa56'),
   '17': ObjectId('6ab361d683d0a67233feaa57'),
   '18': ObjectId('6ab361d683d0a67233feaa58'),
   '19': ObjectId('6ab361d683d0a67233feaa59'),
   '20': ObjectId('6ab361d683d0a67233feaa5a'),
   '21': ObjectId('6ab361d683d0a67233feaa5b'),
   '22': ObjectId('6ab361d683d0a67233feaa5c'),
   '23': ObjectId('6ab361d683d0a67233feaa5d'),
   '24': ObjectId('6ab361d683d0a67233feaa5e'),
   '25': ObjectId('6ab361d683d0a67233feaa5f'),
   '26': ObjectId('6ab361d683d0a67233feaa60'),
   '27': ObjectId('6ab361d683d0a67233feaa61'),
   '28': ObjectId('6ab361d683d0a67233feaa62'),
   '29': ObjectId('6ab361d683d0a67233feaa63')
 }
}
```

4.DISPLAY PRODUCTS

MongoDB Query:

➤ `db.Products.Find()`

```
>_MONGOSH
> db.Products.find()
< {
  _id: ObjectId('6ab361d683d0a67233feaa46'),
  product_id: 'P001',
  name: 'Wireless Bluetooth Headphones',
  category: 'Electronics',
  brand: 'Sony',
  price: 2499,
  rating: 4.5,
  stock: 25,
  seller: 'TechStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa47'),
  product_id: 'P002',
  name: 'Smart Watch',
  category: 'Electronics',
  brand: 'boAt',
  price: 1999,
  rating: 4.2,
  stock: 40,
  seller: 'GadgetWorld'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa48'),
  product_id: 'P003',
  name: 'Running Shoes',
  category: 'Footwear',
  brand: 'Nike',
  price: 3499,
  rating: 4.6,
  stock: 18,
  seller: 'SportsHub'
}
```

➤ `db.Products.FindOne()`

```
>_MONGOSH
> db.Products.findOne()
< {
  _id: ObjectId('6ab361d683d0a67233feaa46'),
  product_id: 'P001',
  name: 'Wireless Bluetooth Headphones',
  category: 'Electronics',
  brand: 'Sony',
  price: 2499,
  rating: 4.5,
  stock: 25,
  seller: 'TechStore'
}
```

5.DISPLAY PRODUCTS BY A FIELD

MongoDB Query:

➤ `db.Products.find({category: "Electronics"})`

```
>_MONGOSH
> db.Products.find({category: "Electronics"})
< {
  _id: ObjectId('6ab361d683d0a67233feaa46'),
  product_id: 'P001',
  name: 'Wireless Bluetooth Headphones',
  category: 'Electronics',
  brand: 'Sony',
  price: 2499,
  rating: 4.5,
  stock: 25,
  seller: 'TechStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa47'),
  product_id: 'P002',
  name: 'Smart Watch',
  category: 'Electronics',
  brand: 'boAt',
  price: 1999,
  rating: 4.2,
  stock: 40,
  seller: 'GadgetWorld'
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  product_id: 'P006',
  name: 'Mechanical Keyboard',
  category: 'Electronics',
  brand: 'Redragon',
  price: 2899,
  rating: 4.5,
  stock: 15,
  seller: 'TechStore'
}
```

```
>_MONGOSH
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4c'),
  product_id: 'P007',
  name: 'USB-C Fast Charger',
  category: 'Electronics',
  brand: 'Samsung',
  price: 1299,
  rating: 4.4,
  stock: 45,
  seller: 'MobileZone'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4f'),
  product_id: 'P010',
  name: 'Portable Bluetooth Speaker',
  category: 'Electronics',
  brand: 'JBL',
  price: 3299,
  rating: 4.7,
  stock: 22,
  seller: 'SoundWorld'
}
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  _id: ObjectId('6ab361d683d0a67233feaa52'),
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  name: 'Gaming Mouse',
  category: 'Electronics',
  brand: 'Logitech',
  price: 1599,
  rating: 4.6,
  stock: 35,
  seller: 'GameZone'
}
```

>_MONGOSH

```
}
{
  _id: ObjectId('6ab361d683d0a67233feaa56'),
  product_id: 'P017',
  name: 'Smartphone',
  category: 'Electronics',
  brand: 'OnePlus',
  price: 29999,
  rating: 4.5,
  stock: 14,
  seller: 'MobileZone'
}
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  _id: ObjectId('6ab361d683d0a67233feaa5d'),
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  brand: 'HP',
  price: 899,
  rating: 4.2,
  stock: 55,
  seller: 'TechStore'
}
```

```
seller: 'TechStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa60'),
  product_id: 'P027',
  name: 'Bluetooth Earbuds',
  category: 'Electronics',
  brand: 'Realme',
  price: 1799,
  rating: 4.3,
  stock: 42,
  seller: 'MobileZone'
}
```

MongoDB Query:

- `db.Products.findOne({product_id: "P020"})`
- `db.Products.findOne({brand: "Sony"})`

>_MONGOSH

```
> db.Products.findOne({product_id: "P020"})
< {
  _id: ObjectId('6ab361d683d0a67233feaa59'),
  product_id: 'P020',
  name: 'Gaming Headset',
  category: 'Electronics',
  brand: 'HyperX',
  price: 4999,
  rating: 4.7,
  stock: 16,
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}
> db.Products.findOne({brand: "Sony"})
< {
  _id: ObjectId('6ab361d683d0a67233feaa46'),
  product_id: 'P001',
  name: 'Wireless Bluetooth Headphones',
  category: 'Electronics',
  brand: 'Sony',
  price: 2499,
  rating: 4.5,
  stock: 25,
  seller: 'TechStore'
}
```

6.DISPLAY PRODUCTS USING COMPARISON OPERATOR

MongoDB Query:

1) Using \$gt

➤ `db.Products.find({price: {$gt: 2000}})`

```
>_MONGOSH
> db.Products.find({price: {$gt: 2000}})
< {
  _id: ObjectId('6ab361d683d0a67233feaa46'),
  product_id: 'P001',
  name: 'Wireless Bluetooth Headphones',
  category: 'Electronics',
  brand: 'Sony',
  price: 2499,
  rating: 4.5,
  stock: 25,
  seller: 'TechStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa48'),
  product_id: 'P003',
  name: 'Running Shoes',
  category: 'Footwear',
  brand: 'Nike',
  price: 3499,
  rating: 4.6,
  stock: 18,
  seller: 'SportsHub'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4b'),
  product_id: 'P006',
  name: 'Mechanical Keyboard',
  category: 'Electronics',
  brand: 'Redragon',
  price: 2899,
  rating: 4.5,
  stock: 15,
  seller: 'TechStore'
}
```

>_MONGOSH



```
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4d'),
  product_id: 'P008',
  name: 'Denim Jacket',
  category: 'Fashion',
  brand: 'Levi's',
  price: 2799,
  rating: 4.5,
  stock: 20,
  seller: 'FashionHub'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4f'),
  product_id: 'P010',
  name: 'Portable Bluetooth Speaker',
  category: 'Electronics',
  brand: 'JBL',
  price: 3299,
  rating: 4.7,
  stock: 22,
  seller: 'SoundWorld'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa51'),
  product_id: 'P012',
  name: 'Coffee Maker',
  category: 'Home & Kitchen',
  brand: 'Philips',
  price: 3499,
  rating: 4.4,
  stock: 12,
  seller: 'HomeStore'
```

>_MONGOSH



```
}
{
  _id: ObjectId('6ab361d683d0a67233feaa55'),
  product_id: 'P016',
  name: 'Air Fryer',
  category: 'Home & Kitchen',
  brand: 'Philips',
  price: 6999,
  rating: 4.6,
  stock: 10,
  seller: 'HomeStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa56'),
  product_id: 'P017',
  name: 'Smartphone',
  category: 'Electronics',
  brand: 'OnePlus',
  price: 29999,
  rating: 4.5,
  stock: 14,
  seller: 'MobileZone'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa59'),
  product_id: 'P020',
  name: 'Gaming Headset',
  category: 'Electronics',
  brand: 'HyperX',
  price: 4999,
  rating: 4.7,
  stock: 16,
  seller: 'GameZone'
```

```

    seller: 'GameZone'
  }
  {
    _id: ObjectId('6ab361d683d0a67233feaa5a'),
    product_id: 'P021',
    name: 'Skin Care Gift Set',
    category: 'Beauty',
    brand: 'The Body Shop',
    price: 2199,
    rating: 4.4,
    stock: 24,
    seller: 'BeautyMart'
  }
  {
    _id: ObjectId('6ab361d683d0a67233feaa5c'),
    product_id: 'P023',
    name: 'Cricket Bat',
    category: 'Sports',
    brand: 'SG',
    price: 2499,
    rating: 4.5,
    stock: 17,
    seller: 'SportsHub'
  }
}

```

2) Using \$lt

➤ `db.Products.find({price: {$lt: 1000}})`

```

p_MONGOSH
> db.Products.find({price: {$lt: 1000}})
< {
  _id: ObjectId('6ab361d683d0a67233feaa49'),
  product_id: 'P004',
  name: 'Face Moisturizer',
  category: 'Beauty',
  brand: 'Cetaphil',
  price: 599,
  rating: 4.4,
  stock: 50,
  seller: 'BeautyMart'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4e'),
  product_id: 'P009',
  name: 'Stainless Steel Water Bottle',
  category: 'Home & Kitchen',
  brand: 'Milton',
  price: 699,
  rating: 4.3,
  stock: 60,
  seller: 'HomeStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa50'),
  product_id: 'P011',
  name: 'Cotton T-Shirt',
  category: 'Fashion',
  brand: 'H&M',
  price: 799,
  rating: 4.1,
  stock: 75,
  seller: 'FashionHub'
}

```

_MONGOSH



```
}
{
  _id: ObjectId('6ab361d683d0a67233feaa53'),
  product_id: 'P014',
  name: 'Sunscreen SPF 50',
  category: 'Beauty',
  brand: 'Minimalist',
  price: 499,
  rating: 4.5,
  stock: 80,
  seller: 'BeautyMart'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa54'),
  product_id: 'P015',
  name: 'Yoga Mat',
  category: 'Sports',
  brand: 'Boldfit',
  price: 899,
  rating: 4.2,
  stock: 40,
  seller: 'SportsHub'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa5d'),
  product_id: 'P024',
  name: 'Wireless Mouse',
  category: 'Electronics',
  brand: 'HP',
  price: 899,
  rating: 4.2,
  stock: 55,
  seller: 'TechStore'
}
```

```
}
{
  _id: ObjectId('6ab361d683d0a67233feaa62'),
  product_id: 'P029',
  name: 'Foundation Makeup',
  category: 'Beauty',
  brand: 'Maybelline',
  price: 749,
  rating: 4.2,
  stock: 48,
  seller: 'BeautyMart'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa63'),
  product_id: 'P030',
  name: 'Resistance Bands Set',
  category: 'Sports',
  brand: 'Decathlon',
  price: 699,
  rating: 4.4,
  stock: 65,
  seller: 'SportsHub'
}
}
```

3) Using \$gte

➤ `db.Products.find({price: {$gte: 2000}})`

```
_MONGOSH
> db.Products.find({price: {$gte: 2000}})
< {
  _id: ObjectId('6ab361d683d0a67233feaa46'),
  product_id: 'P001',
  name: 'Wireless Bluetooth Headphones',
  category: 'Electronics',
  brand: 'Sony',
  price: 2499,
  rating: 4.5,
  stock: 25,
  seller: 'TechStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa48'),
  product_id: 'P003',
  name: 'Running Shoes',
  category: 'Footwear',
  brand: 'Nike',
  price: 3499,
  rating: 4.6,
  stock: 18,
  seller: 'SportsHub'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4b'),
  product_id: 'P006',
  name: 'Mechanical Keyboard',
  category: 'Electronics',
  brand: 'Redragon',
  price: 2899,
  rating: 4.5,
  stock: 15,
  seller: 'TechStore'
}
```

```
>_MONGOSH
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4d'),
  product_id: 'P008',
  name: 'Denim Jacket',
  category: 'Fashion',
  brand: "Levi's",
  price: 2799,
  rating: 4.5,
  stock: 20,
  seller: 'FashionHub'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4f'),
  product_id: 'P010',
  name: 'Portable Bluetooth Speaker',
  category: 'Electronics',
  brand: 'JBL',
  price: 3299,
  rating: 4.7,
  stock: 22,
  seller: 'SoundWorld'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa51'),
  product_id: 'P012',
  name: 'Coffee Maker',
  category: 'Home & Kitchen',
  brand: 'Philips',
  price: 3499,
  rating: 4.4,
  stock: 12,
  seller: 'HomeStore'
}
```

```
}
{
  _id: ObjectId('6ab361d683d0a67233feaa55'),
  product_id: 'P016',
  name: 'Air Fryer',
  category: 'Home & Kitchen',
  brand: 'Philips',
  price: 6999,
  rating: 4.6,
  stock: 10,
  seller: 'HomeStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa56'),
  product_id: 'P017',
  name: 'Smartphone',
  category: 'Electronics',
  brand: 'OnePlus',
  price: 29999,
  rating: 4.5,
  stock: 14,
  seller: 'MobileZone'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa59'),
  product_id: 'P020',
  name: 'Gaming Headset',
  category: 'Electronics',
  brand: 'HyperX',
  price: 4999,
  rating: 4.7,
  stock: 16,
  seller: 'GameZone'
}
```

```
{
  _id: ObjectId('6ab361d683d0a67233feaa5a'),
  product_id: 'P021',
  name: 'Skin Care Gift Set',
  category: 'Beauty',
  brand: 'The Body Shop',
  price: 2199,
  rating: 4.4,
  stock: 24,
  seller: 'BeautyMart'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa5c'),
  product_id: 'P023',
  name: 'Cricket Bat',
  category: 'Sports',
  brand: 'SG',
  price: 2499,
  rating: 4.5,
  stock: 17,
  seller: 'SportsHub'
}
```

4) Using \$lte

➤ `db.Products.find({price: {$lte: 1000}})`

```
_MONGOSH
> db.Products.find({price: {$lte: 1000}})
< {
  _id: ObjectId('6ab361d683d0a67233feaa49'),
  product_id: 'P004',
  name: 'Face Moisturizer',
  category: 'Beauty',
  brand: 'Cetaphil',
  price: 599,
  rating: 4.4,
  stock: 50,
  seller: 'BeautyMart'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4e'),
  product_id: 'P009',
  name: 'Stainless Steel Water Bottle',
  category: 'Home & Kitchen',
  brand: 'Milton',
  price: 699,
  rating: 4.3,
  stock: 60,
  seller: 'HomeStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa50'),
  product_id: 'P011',
  name: 'Cotton T-Shirt',
  category: 'Fashion',
  brand: 'H&M',
  price: 799,
  rating: 4.1,
  stock: 75,
  seller: 'FashionHub'
}
```

```
_MONGOSH
}
{
  _id: ObjectId('6ab361d683d0a67233feaa53'),
  product_id: 'P014',
  name: 'Sunscreen SPF 50',
  category: 'Beauty',
  brand: 'Minimalist',
  price: 499,
  rating: 4.5,
  stock: 80,
  seller: 'BeautyMart'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa54'),
  product_id: 'P015',
  name: 'Yoga Mat',
  category: 'Sports',
  brand: 'Boldfit',
  price: 899,
  rating: 4.2,
  stock: 40,
  seller: 'SportsHub'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa5d'),
  product_id: 'P024',
  name: 'Wireless Mouse',
  category: 'Electronics',
  brand: 'HP',
  price: 899,
  rating: 4.2,
  stock: 55,
  seller: 'TechStore'
}
```

```

    {
      _id: ObjectId('6ab361d683d0a67233feaa62'),
      product_id: 'P029',
      name: 'Foundation Makeup',
      category: 'Beauty',
      brand: 'Maybelline',
      price: 749,
      rating: 4.2,
      stock: 48,
      seller: 'BeautyMart'
    }
  ],
  {
    _id: ObjectId('6ab361d683d0a67233feaa63'),
    product_id: 'P030',
    name: 'Resistance Bands Set',
    category: 'Sports',
    brand: 'Decathlon',
    price: 699,
    rating: 4.4,
    stock: 65,
    seller: 'SportsHub'
  }
]

```

5) Using \$ne

➤ `db.Products.find({category:{$ne:"Electronics"}})`

```

> _MONGOOSH
> db.Products.findOne({category: {$ne: "Electronics"}})
< {
  _id: ObjectId('6ab361d683d0a67233feaa48'),
  product_id: 'P003',
  name: 'Running Shoes',
  category: 'Footwear',
  brand: 'Nike',
  price: 3499,
  rating: 4.6,
  stock: 18,
  seller: 'SportsHub'
}
> db.Products.find({category: {$ne: "Electronics"}})
< [
  {
    _id: ObjectId('6ab361d683d0a67233feaa48'),
    product_id: 'P003',
    name: 'Running Shoes',
    category: 'Footwear',
    brand: 'Nike',
    price: 3499,
    rating: 4.6,
    stock: 18,
    seller: 'SportsHub'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa49'),
    product_id: 'P004',
    name: 'Face Moisturizer',
    category: 'Beauty',
    brand: 'Cetaphil',
    price: 599,
    rating: 4.4,
    stock: 50,

```

```
{
  _id: ObjectId('6ab361d683d0a67233feaa62'),
  product_id: 'P029',
  name: 'Foundation Makeup',
  category: 'Beauty',
  brand: 'Maybelline',
  price: 749,
  rating: 4.2,
  stock: 48,
  seller: 'BeautyMart'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa63'),
  product_id: 'P030',
  name: 'Resistance Bands Set',
  category: 'Sports',
  brand: 'Decathlon',
  price: 699,
  rating: 4.4,
  stock: 65,
```

```
seller: 'SportsHub'
}
```

7.DISPLAY PRODUCTS BETWEEN A RANGE

MongoDB Query:

➤ `db.Products.find({price: {$gt: 1000, $lt: 3000}})`

```
_MONGOSH
> db.Products.find({price: {$gte: 1000, $lte: 3000}})
{
  _id: ObjectId('6ab361d683d0a67233feaa46'),
  product_id: 'P001',
  name: 'Wireless Bluetooth Headphones',
  category: 'Electronics',
  brand: 'Sony',
  price: 2499,
  rating: 4.5,
  stock: 25,
  seller: 'TechStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa47'),
  product_id: 'P002',
  name: 'Smart Watch',
  category: 'Electronics',
  brand: 'boAt',
  price: 1999,
  rating: 4.2,
  stock: 40,
  seller: 'GadgetWorld'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4a'),
  product_id: 'P005',
  name: 'Laptop Backpack',
  category: 'Accessories',
  brand: 'American Tourister',
  price: 1299,
  rating: 4.3,
  stock: 30,
  seller: 'BagWorld'
}
```

```

}
{
  _id: ObjectId('6ab361d683d0a67233feaa61'),
  product_id: 'P028',
  name: 'Backpack',
  category: 'Accessories',
  brand: 'Wildcraft',
  price: 1599,
  rating: 4.5,
  stock: 31,
  seller: 'BagWorld'
}

```

8.DISPLAY PRODUCTS USING LOGICAL OPERATORS

MongoDB Query:

1)AND

➤ db.Products.find({\$and: [{category: "Electronics"}, {price: {\$lt: 3000}}]})

```

s_MONGOSH
> db.Products.find({$and: [{category: "Electronics"}, {price: {$lt: 3000}}]})
< [
  {
    _id: ObjectId('6ab361d683d0a67233feaa46'),
    product_id: 'P001',
    name: 'Wireless Bluetooth Headphones',
    category: 'Electronics',
    brand: 'Sony',
    price: 2499,
    rating: 4.5,
    stock: 25,
    seller: 'TechStore'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa47'),
    product_id: 'P002',
    name: 'Smart Watch',
    category: 'Electronics',
    brand: 'boAt',
    price: 1999,
    rating: 4.2,
    stock: 40,
    seller: 'GadgetWorld'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa4b'),
    product_id: 'P006',
    name: 'Mechanical Keyboard',
    category: 'Electronics',
    brand: 'Redragon',
    price: 2899,
    rating: 4.5,
    stock: 15,
    seller: 'TechStore'
  }
]

```

```
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4c'),
  product_id: 'P007',
  name: 'USB-C Fast Charger',
  category: 'Electronics',
  brand: 'Samsung',
  price: 1299,
  rating: 4.4,
  stock: 45,
  seller: 'MobileZone'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa52'),
  product_id: 'P013',
  name: 'Gaming Mouse',
  category: 'Electronics',
  brand: 'Logitech',
  price: 1599,
  rating: 4.6,
  stock: 35,
  seller: 'GameZone'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa5d'),
  product_id: 'P024',
  name: 'Wireless Mouse',
  category: 'Electronics',
  brand: 'HP',
  price: 899,
  rating: 4.2,
  stock: 55,
  seller: 'TechStore'
}
```

```
seller: 'TechStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa60'),
  product_id: 'P027',
  name: 'Bluetooth Earbuds',
  category: 'Electronics',
  brand: 'Realme',
  price: 1799,
  rating: 4.3,
  stock: 42,
  seller: 'MobileZone'
}
```

2) OR

➤ `db.Products.find({$or: [{brand: "Sony"}, {brand: "JBL"}]})`

```
> db.Products.find({$or: [{brand: "Sony"}, {brand: "JBL"}]})
< {
  _id: ObjectId('6ab361d683d0a67233feaa46'),
  product_id: 'P001',
  name: 'Wireless Bluetooth Headphones',
  category: 'Electronics',
  brand: 'Sony',
  price: 2499,
  rating: 4.5,
  stock: 25,
  seller: 'TechStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4f'),
  product_id: 'P010',
  name: 'Portable Bluetooth Speaker',
  category: 'Electronics',
  brand: 'JBL',
  price: 3299,
  rating: 4.7,
  stock: 22,
  seller: 'SoundWorld'
}
```

3) NOR

➤ `db.Products.find({$nor: [{seller: "BeautyMart"}, {category: "Electronics"}]})`

```
> _MONGOSH
> db.Products.find({$nor: [{seller: "BeautyMart"}, {category: "Electronics"}]})
< [
  {
    _id: ObjectId('6ab361d683d0a67233feaa48'),
    product_id: 'P003',
    name: 'Running Shoes',
    category: 'Footwear',
    brand: 'Nike',
    price: 3499,
    rating: 4.6,
    stock: 18,
    seller: 'SportsHub'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa4a'),
    product_id: 'P005',
    name: 'Laptop Backpack',
    category: 'Accessories',
    brand: 'American Tourister',
    price: 1299,
    rating: 4.3,
    stock: 30,
    seller: 'BagWorld'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa4d'),
    product_id: 'P008',
    name: 'Denim Jacket',
    category: 'Fashion',
    brand: 'Levi's',
    price: 2799,
    rating: 4.5,
    stock: 20,
    seller: 'FashionHub'
  }
]
```

```
}
{
  _id: ObjectId('6ab361d683d0a67233feaa63'),
  product_id: 'P030',
  name: 'Resistance Bands Set',
  category: 'Sports',
  brand: 'Decathlon',
  price: 699,
  rating: 4.4,
  stock: 65,
  seller: 'SportsHub'
}
```

4)NOT

➤ `db.Products.find({price: {$not: {$gt: 1000}}})`

```
>_MONGOSH
> db.Products.find({price: {$not: {$gt: 1000}}})
< [
  {
    _id: ObjectId('6ab361d683d0a67233feaa49'),
    product_id: 'P004',
    name: 'Face Moisturizer',
    category: 'Beauty',
    brand: 'Cetaphil',
    price: 599,
    rating: 4.4,
    stock: 50,
    seller: 'BeautyMart'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa4e'),
    product_id: 'P009',
    name: 'Stainless Steel Water Bottle',
    category: 'Home & Kitchen',
    brand: 'Milton',
    price: 699,
    rating: 4.3,
    stock: 60,
    seller: 'HomeStore'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa50'),
    product_id: 'P011',
    name: 'Cotton T-Shirt',
    category: 'Fashion',
    brand: 'H&M',
    price: 799,
    rating: 4.1,
    stock: 75,
    seller: 'FashionHub'
  }
]
```

```
_MONGOSH
}
[
  {
    _id: ObjectId('6ab361d683d0a67233feaa53'),
    product_id: 'P014',
    name: 'Sunscreen SPF 50',
    category: 'Beauty',
    brand: 'Minimalist',
    price: 499,
    rating: 4.5,
    stock: 80,
    seller: 'BeautyMart'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa54'),
    product_id: 'P015',
    name: 'Yoga Mat',
    category: 'Sports',
    brand: 'Boldfit',
    price: 899,
    rating: 4.2,
    stock: 40,
    seller: 'SportsHub'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa5d'),
    product_id: 'P024',
    name: 'Wireless Mouse',
    category: 'Electronics',
    brand: 'HP',
    price: 899,
    rating: 4.2,
    stock: 55,
    seller: 'TechStore'
  }
]
```

```

}
{
  _id: ObjectId('6ab361d683d0a67233feaa62'),
  product_id: 'P029',
  name: 'Foundation Makeup',
  category: 'Beauty',
  brand: 'Maybelline',
  price: 749,
  rating: 4.2,
  stock: 48,
  seller: 'BeautyMart'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa63'),
  product_id: 'P030',
  name: 'Resistance Bands Set',
  category: 'Sports',
  brand: 'Decathlon',
  price: 699,
  rating: 4.4,
  stock: 65,
  seller: 'SportsHub'
}
}

```

9.DISPLAY PRODUCTS USING MEMBERSHIP OPERATOR

MongoDB Query:

1) IN

➤ `db.Products.find({brand: {$in: ["Sony", "JBL", "Logitech"]}})`

```

_MONGOSH
> db.Products.find({brand: {$in: ["Sony", "JBL", "Logitech"]}})
< {
  _id: ObjectId('6ab361d683d0a67233feaa46'),
  product_id: 'P001',
  name: 'Wireless Bluetooth Headphones',
  category: 'Electronics',
  brand: 'Sony',
  price: 2499,
  rating: 4.5,
  stock: 25,
  seller: 'TechStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4f'),
  product_id: 'P010',
  name: 'Portable Bluetooth Speaker',
  category: 'Electronics',
  brand: 'JBL',
  price: 3299,
  rating: 4.7,
  stock: 22,
  seller: 'SoundWorld'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa52'),
  product_id: 'P013',
  name: 'Gaming Mouse',
  category: 'Electronics',
  brand: 'Logitech',
  price: 1599,
  rating: 4.6,
  stock: 35,
  seller: 'GameZone'
}

```

2) NOT IN

➤ `db.Products.find({category: {$nin: ["Electronics", "Beauty", "Accessories", "Sports"]}})`

```
_MONGOSH
> db.Products.find({category: {$nin: ["Electronics", "Beauty", "Accessories", "Sports"]}})
< [
  {
    _id: ObjectId('6ab361d683d0a67233feaa48'),
    product_id: 'P003',
    name: 'Running Shoes',
    category: 'Footwear',
    brand: 'Nike',
    price: 3499,
    rating: 4.6,
    stock: 18,
    seller: 'SportsHub'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa4d'),
    product_id: 'P008',
    name: 'Denim Jacket',
    category: 'Fashion',
    brand: 'Levi's',
    price: 2799,
    rating: 4.5,
    stock: 20,
    seller: 'FashionHub'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa4e'),
    product_id: 'P009',
    name: 'Stainless Steel Water Bottle',
    category: 'Home & Kitchen',
    brand: 'Milton',
    price: 699,
    rating: 4.3,
    stock: 60,
    seller: 'HomeStore'
  }
]
```

```
_MONGOSH
}
[
  {
    _id: ObjectId('6ab361d683d0a67233feaa50'),
    product_id: 'P011',
    name: 'Cotton T-Shirt',
    category: 'Fashion',
    brand: 'H&M',
    price: 799,
    rating: 4.1,
    stock: 75,
    seller: 'FashionHub'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa51'),
    product_id: 'P012',
    name: 'Coffee Maker',
    category: 'Home & Kitchen',
    brand: 'Philips',
    price: 3499,
    rating: 4.4,
    stock: 12,
    seller: 'HomeStore'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa55'),
    product_id: 'P016',
    name: 'Air Fryer',
    category: 'Home & Kitchen',
    brand: 'Philips',
    price: 6999,
    rating: 4.6,
    stock: 10,
    seller: 'HomeStore'
  }
]
```

```
>_MONGOSH
}
{
  _id: ObjectId('6ab361d683d0a67233feaa58'),
  product_id: 'P019',
  name: 'LED Desk Lamp',
  category: 'Home & Kitchen',
  brand: 'Wipro',
  price: 1099,
  rating: 4.3,
  stock: 32,
  seller: 'HomeStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa5b'),
  product_id: 'P022',
  name: 'Formal Shirt',
  category: 'Fashion',
  brand: 'Van Heusen',
  price: 1599,
  rating: 4.1,
  stock: 35,
  seller: 'FashionHub'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa5e'),
  product_id: 'P025',
  name: 'Electric Kettle',
  category: 'Home & Kitchen',
  brand: 'Prestige',
  price: 1499,
  rating: 4.3,
  stock: 26,
  seller: 'HomeStore'
}
```

10.DISPLAY PRODUCTS USING SORT ()

MongoDB Query:

➤ `db.Products.find().sort({price: 1})`

```
> db.Products.find().sort({price: 1})
< {
  _id: ObjectId('6ab361d683d0a67233feaa53'),
  product_id: 'P014',
  name: 'Sunscreen SPF 50',
  category: 'Beauty',
  brand: 'Minimalist',
  price: 499,
  rating: 4.5,
  stock: 80,
  seller: 'BeautyMart'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa49'),
  product_id: 'P004',
  name: 'Face Moisturizer',
  category: 'Beauty',
  brand: 'Cetaphil',
  price: 599,
  rating: 4.4,
  stock: 50,
  seller: 'BeautyMart'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4e'),
  product_id: 'P009',
  name: 'Stainless Steel Water Bottle',
  category: 'Home & Kitchen',
  brand: 'Milton',
  price: 699,
  rating: 4.3,
  stock: 40,
  seller: 'HomeStore'
}
```

```

>_MONGOSH
    stock: 60,
    seller: 'HomeStore'
  }
  {
    _id: ObjectId('6ab361d683d0a67233feaa63'),
    product_id: 'P030',
    name: 'Resistance Bands Set',
    category: 'Sports',
    brand: 'Decathlon',
    price: 699,
    rating: 4.4,
    stock: 65,
    seller: 'SportsHub'
  }
  {
    _id: ObjectId('6ab361d683d0a67233feaa62'),
    product_id: 'P029',
    name: 'Foundation Makeup',
    category: 'Beauty',
    brand: 'Maybelline',
    price: 749,
    rating: 4.2,
    stock: 48,
    seller: 'BeautyMart'
  }
}

```

➤ `db.Products.find().sort({price: -1})`

```

>_MONGOSH
> db.Products.find().sort({price: -1})
< {
  _id: ObjectId('6ab361d683d0a67233feaa56'),
  product_id: 'P017',
  name: 'Smartphone',
  category: 'Electronics',
  brand: 'OnePlus',
  price: 29999,
  rating: 4.5,
  stock: 14,
  seller: 'MobileZone'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa55'),
  product_id: 'P016',
  name: 'Air Fryer',
  category: 'Home & Kitchen',
  brand: 'Philips',
  price: 6999,
  rating: 4.6,
  stock: 10,
  seller: 'HomeStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa59'),
  product_id: 'P020',
  name: 'Gaming Headset',
  category: 'Electronics',
  brand: 'HyperX',
  price: 4999,
  rating: 4.7,
  stock: 16,
  seller: 'GameZone'
}

```

```

>_MONGOSH
}
{
  _id: ObjectId('6ab361d683d0a67233feaa48'),
  product_id: 'P003',
  name: 'Running Shoes',
  category: 'Footwear',
  brand: 'Nike',
  price: 3499,
  rating: 4.6,
  stock: 18,
  seller: 'SportsHub'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa51'),
  product_id: 'P012',
  name: 'Coffee Maker',
  category: 'Home & Kitchen',
  brand: 'Philips',
  price: 3499,
  rating: 4.4,
  stock: 12,
  seller: 'HomeStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa4f'),
  product_id: 'P010',
  name: 'Portable Bluetooth Speaker',
  category: 'Electronics',
  brand: 'JBL',
  price: 3299,
  rating: 4.7,
  stock: 22,
  seller: 'SoundWorld'
}

```

11.DISPLAY PRODUCTS USING LIMIT ()

MongoDB Query:

➤ `db.Products.find().limit(5)`

```

>_MONGOSH
> db.Products.find().limit(5)
< {
  _id: ObjectId('6ab361d683d0a67233feaa46'),
  product_id: 'P001',
  name: 'Wireless Bluetooth Headphones',
  category: 'Electronics',
  brand: 'Sony',
  price: 2499,
  rating: 4.5,
  stock: 25,
  seller: 'TechStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa47'),
  product_id: 'P002',
  name: 'Smart Watch',
  category: 'Electronics',
  brand: 'boAt',
  price: 1999,
  rating: 4.2,
  stock: 40,
  seller: 'GadgetWorld'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa48'),
  product_id: 'P003',
  name: 'Running Shoes',
  category: 'Footwear',
  brand: 'Nike',
  price: 3499,
  rating: 4.6,
  stock: 18,
  seller: 'SportsHub'
}

```

> db.Products.find().sort({price: -1}).limit(5)

```
>_MONGOSH
> db.Products.find().sort({price: -1}).limit(5)
< [
  {
    _id: ObjectId('6ab361d683d0a67233feaa56'),
    product_id: 'P017',
    name: 'Smartphone',
    category: 'Electronics',
    brand: 'OnePlus',
    price: 29999,
    rating: 4.5,
    stock: 14,
    seller: 'MobileZone'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa55'),
    product_id: 'P016',
    name: 'Air Fryer',
    category: 'Home & Kitchen',
    brand: 'Philips',
    price: 6999,
    rating: 4.6,
    stock: 10,
    seller: 'HomeStore'
  },
  {
    _id: ObjectId('6ab361d683d0a67233feaa59'),
    product_id: 'P020',
    name: 'Gaming Headset',
    category: 'Electronics',
    brand: 'HyperX',
    price: 4999,
    rating: 4.7,
    stock: 16,
    seller: 'GameZone'
  }
]
```

```

}
{
  _id: ObjectId('6ab361d683d0a67233feaa51'),
  product_id: 'P012',
  name: 'Coffee Maker',
  category: 'Home & Kitchen',
  brand: 'Philips',
  price: 3499,
  rating: 4.4,
  stock: 12,
  seller: 'HomeStore'
}
{
  _id: ObjectId('6ab361d683d0a67233feaa48'),
  product_id: 'P003',
  name: 'Running Shoes',
  category: 'Footwear',
  brand: 'Nike',
  price: 3499,
  rating: 4.6,
  stock: 18,
  seller: 'SportsHub'
}
}
```

12.DISPLAY THE TOTAL NUMBER OF PRODUCTS

MongoDB Query:

➤ db.Products.countDocuments()

```
> db.Products.countDocuments()  
< 30
```

➤ db.Products.countDocuments({category: "Electronics"})

➤ db.Products.countDocuments({rating: {\$gt: 4.5}})

➤ db.Products.countDocuments({price: {\$lt: 2000}})

➤ db.Products.countDocuments({seller: "TechStore", rating: {\$gte: 4.5}})

```
> db.Products.countDocuments({category: "Electronics"})  
< 10  
> db.Products.countDocuments({rating: {$gt: 4.5}})  
< 5  
> db.Products.countDocuments({price: {$lt: 2000}})  
< 19  
> db.Products.countDocuments({seller: "TechStore", rating: {$gte: 4.5}})  
< 2
```

13.USING CRUD OPERATION ON

REVIEWS COLLECTION

MongoDB Query:

1)Create

➤ db.Reviews.insertMany([{review_id: “R001”, ...}])

```
> MONGOSH
> use("EcommerceDB")
< switched to db EcommerceDB
> db.Reviews.insertMany([
  { review_id: "R001", product_id: "P001", customer: "Aarav", rating: 5, comment: "Excellent sound quality and comfortable to wear.", verified_purchase: true },
  { review_id: "R002", product_id: "P001", customer: "Riya", rating: 4, comment: "Good headphones but the battery could be better.", verified_purchase: true },
  { review_id: "R003", product_id: "P002", customer: "Kabir", rating: 4, comment: "Useful smartwatch with a nice display.", verified_purchase: true },
  { review_id: "R004", product_id: "P003", customer: "Ananya", rating: 5, comment: "Very comfortable shoes for running.", verified_purchase: true },
  { review_id: "R005", product_id: "P004", customer: "Meera", rating: 4, comment: "Moisturizer feels light and non-sticky.", verified_purchase: false },
  { review_id: "R006", product_id: "P005", customer: "Rahul", rating: 5, comment: "Spacious and durable backpack.", verified_purchase: true },
  { review_id: "R007", product_id: "P006", customer: "Arjun", rating: 4, comment: "Great keyboard with satisfying keys.", verified_purchase: true },
  { review_id: "R008", product_id: "P007", customer: "Ishita", rating: 3, comment: "Charging speed is good but it gets slightly warm.", verified_purchase: true },
  { review_id: "R009", product_id: "P008", customer: "Vivaan", rating: 5, comment: "Looks stylish and fits perfectly.", verified_purchase: false },
  { review_id: "R010", product_id: "P009", customer: "Sneha", rating: 4, comment: "Good quality bottle and keeps water cool.", verified_purchase: true },
  { review_id: "R011", product_id: "P010", customer: "Aditya", rating: 5, comment: "Loud and clear sound. Very good speaker.", verified_purchase: true },
  { review_id: "R012", product_id: "P011", customer: "Diya", rating: 3, comment: "The fabric is comfortable but the fit is average.", verified_purchase: true },
  { review_id: "R013", product_id: "P013", customer: "Karan", rating: 5, comment: "Smooth and responsive gaming mouse.", verified_purchase: true },
  { review_id: "R014", product_id: "P014", customer: "Nisha", rating: 4, comment: "Lightweight sunscreen with no heavy feeling.", verified_purchase: true },
  { review_id: "R015", product_id: "P016", customer: "Manav", rating: 5, comment: "Makes cooking quick and easy.", verified_purchase: true },
  { review_id: "R016", product_id: "P017", customer: "Tanya", rating: 4, comment: "Good performance and camera quality.", verified_purchase: false },
  { review_id: "R017", product_id: "P020", customer: "Yash", rating: 5, comment: "Excellent gaming headset with clear audio.", verified_purchase: true },
  { review_id: "R018", product_id: "P023", customer: "Sahil", rating: 4, comment: "Good cricket bat with solid build quality.", verified_purchase: true },
  { review_id: "R019", product_id: "P027", customer: "Aanya", rating: 3, comment: "Sound quality is decent but the fit could improve.", verified_purchase: true },
  { review_id: "R020", product_id: "P030", customer: "Dev", rating: 5, comment: "Good resistance bands for home workouts.", verified_purchase: true }
])
```

```
EcommerceDB
├── Products
├── Reviews
├── admin
├── config
└── local

< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ab546e6ba9791b118c78619'),
    '1': ObjectId('6ab546e6ba9791b118c7861a'),
    '2': ObjectId('6ab546e6ba9791b118c7861b'),
    '3': ObjectId('6ab546e6ba9791b118c7861c'),
    '4': ObjectId('6ab546e6ba9791b118c7861d'),
    '5': ObjectId('6ab546e6ba9791b118c7861e'),
    '6': ObjectId('6ab546e6ba9791b118c7861f'),
    '7': ObjectId('6ab546e6ba9791b118c78620'),
    '8': ObjectId('6ab546e6ba9791b118c78621'),
    '9': ObjectId('6ab546e6ba9791b118c78622'),
    '10': ObjectId('6ab546e6ba9791b118c78623'),
    '11': ObjectId('6ab546e6ba9791b118c78624'),
    '12': ObjectId('6ab546e6ba9791b118c78625'),
    '13': ObjectId('6ab546e6ba9791b118c78626'),
    '14': ObjectId('6ab546e6ba9791b118c78627'),
    '15': ObjectId('6ab546e6ba9791b118c78628'),
    '16': ObjectId('6ab546e6ba9791b118c78629'),
    '17': ObjectId('6ab546e6ba9791b118c7862a'),
    '18': ObjectId('6ab546e6ba9791b118c7862b'),
    '19': ObjectId('6ab546e6ba9791b118c7862c')
  }
}
```

➤ `db.Reviews.insertOne({review_id: "R021", ...})`

```
> db.Reviews.insertOne({review_id: "R021", product_id: "P005", customer: "Neha", rating: 4, comment: "Good product.", verified_purchase: true})
< {
  acknowledged: true,
  insertedId: ObjectId('6ab547d4ba9791b118c7862d')
}
```

2) Read

➤ `db.Reviews.find()`

```
> _MONGOSH
> db.Reviews.find()
< [
  {
    _id: ObjectId('6ab546e6ba9791b118c78619'),
    review_id: 'R001',
    product_id: 'P001',
    customer: 'Aarav',
    rating: 5,
    comment: 'Excellent sound quality and comfortable to wear.',
    verified_purchase: true
  },
  {
    _id: ObjectId('6ab546e6ba9791b118c7861a'),
    review_id: 'R002',
    product_id: 'P001',
    customer: 'Riya',
    rating: 4,
    comment: 'Good headphones but the battery could be better.',
    verified_purchase: true
  },
  {
    _id: ObjectId('6ab546e6ba9791b118c7861b'),
    review_id: 'R003',
    product_id: 'P002',
    customer: 'Kabir',
    rating: 4,
    comment: 'Useful smartwatch with a nice display.',
    verified_purchase: true
  },
  {
    _id: ObjectId('6ab546e6ba9791b118c7861c'),
    review_id: 'R004',
    product_id: 'P003',
    customer: 'Ananya',

```

.....

```
< {
  _id: ObjectId('6ab547d4ba9791b118c7862d'),
  review_id: 'R021',
  product_id: 'P005',
  customer: 'Neha',
  rating: 4,
  comment: 'Good product.',
  verified_purchase: true
}
```

➤ db.Reviews.findOne()

```
> db.Reviews.findOne()
< {
  _id: ObjectId('6ab546e6ba9791b118c78619'),
  review_id: 'R001',
  product_id: 'P001',
  customer: 'Aarav',
  rating: 5,
  comment: 'Excellent sound quality and comfortable to wear.',
  verified_purchase: true
}
```

3) Update

➤ db.Reviews.updateMany({rating: 3}, {\$set: {verified_purchase: true}})

```
> _MONGOSH
> db.Reviews.updateMany({rating: 3}, {$set: {verified_purchase: true}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 0,
  upsertedCount: 0
}
```

```
> db.Reviews.find({rating: 3})
< [
  {
    _id: ObjectId('6ab546e6ba9791b118c78620'),
    review_id: 'R008',
    product_id: 'P007',
    customer: 'Ishita',
    rating: 3,
    comment: 'Charging speed is good but it gets slightly warm.',
    verified_purchase: true
  },
  {
    _id: ObjectId('6ab546e6ba9791b118c78624'),
    review_id: 'R012',
    product_id: 'P011',
    customer: 'Diya',
    rating: 3,
    comment: 'The fabric is comfortable but the fit is average.',
    verified_purchase: true
  },
  {
    _id: ObjectId('6ab546e6ba9791b118c7862b'),
    review_id: 'R019',
    product_id: 'P027',
    customer: 'Aanya',
    rating: 3,
    comment: 'Sound quality is decent but the fit could improve.',
    verified_purchase: true
  }
]
```

➤ db.Reviews.updateOne({review_id: "R002"},
{ \$set: {rating: 5}}})

```
> db.Reviews.updateOne({review_id: "R002"}, {$set: {rating: 5}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
> db.Reviews.findOne({review_id: "R002"})
< {
  _id: ObjectId('6ab546e6ba9791b118c7861a'),
  review_id: 'R002',
  product_id: 'P001',
  customer: 'Riya',
  rating: 5,
  comment: 'Good headphones but the battery could be better.',
  verified_purchase: true
}
```

3.1) Using \$inc

➤ db.Reviews.updateOne({review_id: "R008"},
{ \$inc: {rating: 1}}})

```
> db.Reviews.updateOne({review_id: "R008"}, {$inc: {rating: 1}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.Reviews.findOne({review_id: "R008"})
< {
  _id: ObjectId('6ab546e6ba9791b118c78620'),
  review_id: 'R008',
  product_id: 'P007',
  customer: 'Ishita',
  rating: 4,
  comment: 'Charging speed is good but it gets slightly warm.',
  verified_purchase: true
}
```

➤ db.Reviews.updateMany({review_id: {\$in: ["R003", "R005", "R018"]}}, {\$inc: {rating: 1}})

```
> db.Reviews.updateMany({review_id: {$in: ["R003", "R005", "R018"]}}, {$inc: {rating: 1}})
```

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

```
> db.Reviews.find({review_id: {$in: ["R003", "R005", "R018"]}})
```

```
< [
  {
    _id: ObjectId('6ab546e6ba9791b118c7861b'),
    review_id: 'R003',
    product_id: 'P002',
    customer: 'Kabir',
    rating: 4,
    comment: 'Useful smartwatch with a nice display.',
    verified_purchase: true
  },
  {
    _id: ObjectId('6ab546e6ba9791b118c7861d'),
    review_id: 'R005',
    product_id: 'P004',
    customer: 'Meera',
    rating: 5,
    comment: 'Moisturizer feels light and non-sticky.',
    verified_purchase: false
  },
  {
    _id: ObjectId('6ab546e6ba9791b118c7862a'),
    review_id: 'R018',
    product_id: 'P023',
    customer: 'Sahil',
    rating: 5,
    comment: 'Good cricket bat with solid build quality.',
    verified_purchase: true
  }
]
```

4) Delete

➤ `db.Reviews.deleteOne({review_id: "R021"})`

```
> db.Reviews.deleteOne({review_id: "R021"})
< {
  acknowledged: true,
  deletedCount: 1
}
> db.Reviews.findOne({review_id: "R021"})
< null
```

➤ `db.Reviews.deleteMany({rating: 3})`

```
> db.Reviews.deleteMany({rating: 3})
< {
  acknowledged: true,
  deletedCount: 2
}
> db.Reviews.find({rating: 3})
<
```